



Erfassung non-verbaler Kommunikation zur Bestimmung der Glasgow Coma Scale (GCS)

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Glasgow Coma Score (GCS)

- Non-verbal response
- Verbal response
- Non-verbal or verbal response

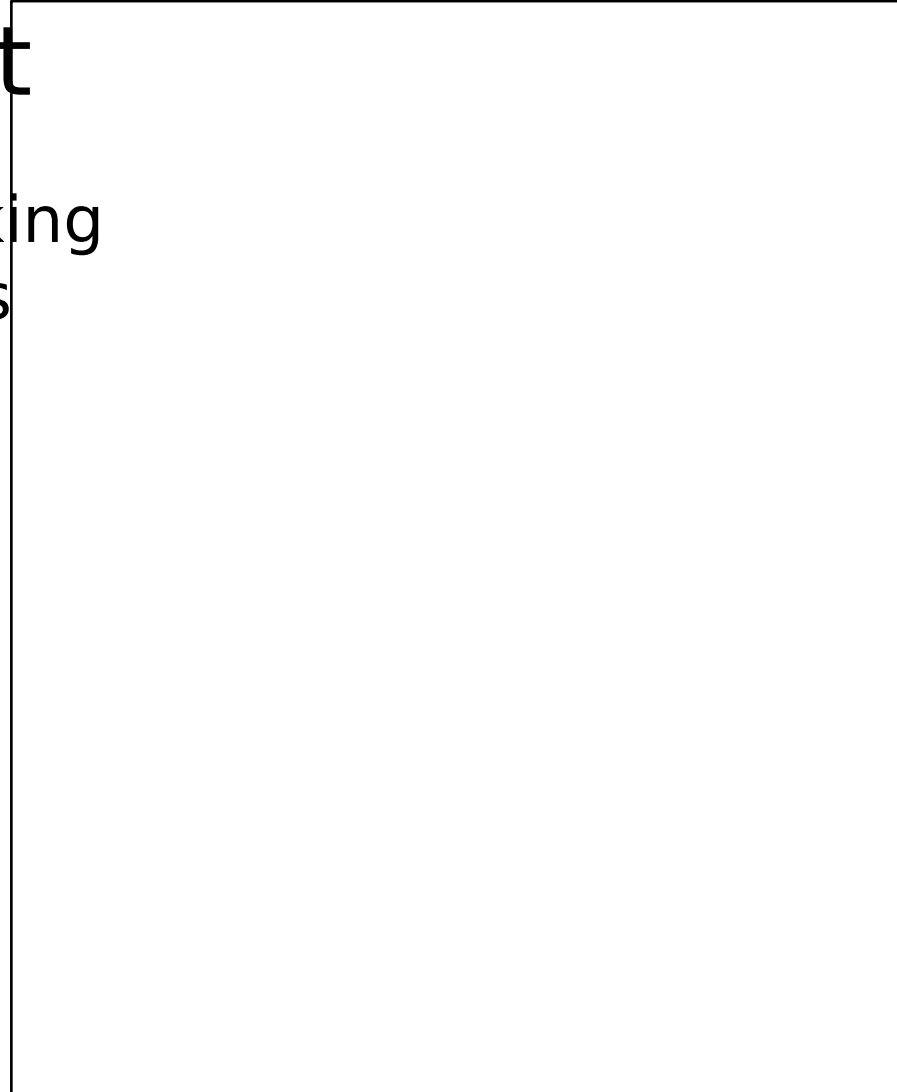
Score	Eye Opening Response	Verbal Response	Motor Response
+6			Obeys commands for movement
+5		Oriented	Purposeful movement to painful ¹ stimulus
+4	Spontaneous-open with blinking at baseline	Confused conversation, but able to answer questions	Withdraws in response to pain ¹
+3	To verbal stimuli, command, speech	Inappropriate words	Flexion in response to pain ¹ (decorticate posturing)
+2	To pain ¹ only (not applied to face)	Incomprehensible speech	Extension response in response to pain ¹ (decerebrate posturing)
	No	No	No

¹ loud noise will be used as pain stimulus

Non-verbal assessment

Combined face and skeleton tracking algorithms provide measurements

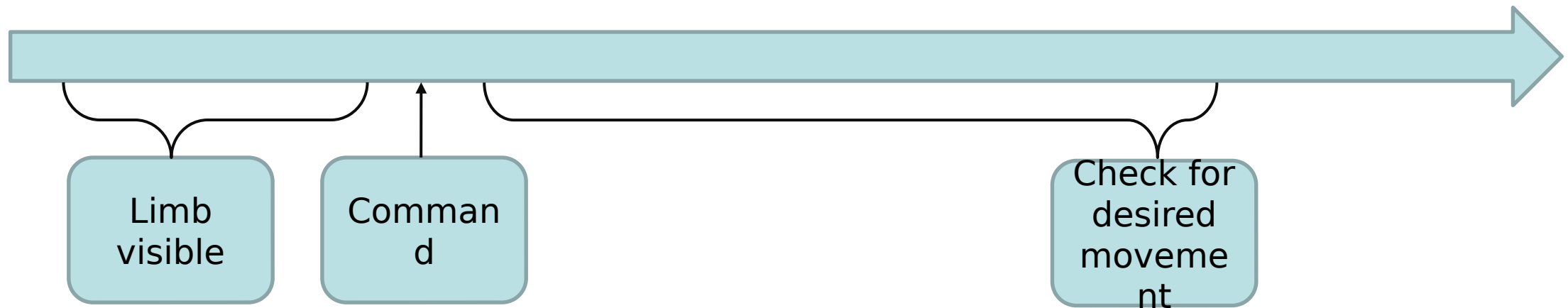
- Eye opening response
 - Eye opening
 - Face orientation
- Verbal response
 - Mouth opening
- Motor response
 - Hand motion



Sensor fusion and semantic classification

- Low-Pass Filtering over time
- Retrospective classification
 - Based on expected behavior, look back some time to evaluate
 - Was the interesting limb even visible beforehand?
 - Obtain semantic values, e.g.:
 - Right arm moved
 - Open mouth

Sensor fusion and semantic classification



User tracking

- In-view tracking
 - Feeding of user object with measured data of the patient
 - Event based structure, link to verbal interaction
 - Starting GCS protocol event when situation is stable
- Face tagging
 - User obtains a persistent face-ID for re-recognition
 - >Object identifier

GCS Module

Dedicated module for computing the Glasgow Coma Score (GCS) based on speech recognition and RGB-D image data analysis

Processor	Intel NUC11TNBi7
Thermal Design Power	28 W
Frequency	4.70 GHz
Cores / Threads	4 / 8
Memory Size	16 GB
Audio co-processor	XMOS XVF-3000
Power / communication	DRZ module slot
Battery backup	20 min

